

Multimodal Temporal Dynamics in Music Genre

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1 Introduction

The thesis investigates how music genres evolve over time across multiple feature modalities. The central idea is that genre evolution should not be described through a single representation only. Instead, the thesis follows a layered analysis strategy that combines dataset-level baselines, centroid-based summary measures, and optional distribution-based track-level representations.

1. **Dataset-level baselines** show how the overall dataset or average genre landscape changes over time. These baselines provide reference points for interpreting selected genre-level trends and distinguishing genre-specific movement from general dataset-wide change.
2. **Centroid-based summaries** describe how the average position of each genre changes over time. They provide the main summary representation for temporal drift and pairwise genre similarity.
3. **Track-level distribution-based analyses** are optional extensions of the centroid-based analyses. While a centroid represents a genre-window using one average vector, distribution-based methods also consider variation between individual tracks. Their scope is summarized in Table 1.
4. **Interpretable raw and normalized feature analyses** connect abstract movement in feature space to concrete acoustic or lyrical changes. For example, if a genre becomes relatively louder in normalized Essentia space, the corresponding raw loudness values in dB and the average dataset baseline will be shown where possible. Possible effects of changes in YouTube normalization, compression, and media quality will be considered in the discussion and limitations.

Comparison	Example	What it measures
Tracks within the same genre and period	Rock tracks within 1980–1984	Internal consistency or diversity of a genre within one period
Same genre across different periods	Rock in 1980–1984 compared with Rock in 2000–2004	Temporal change in the complete track-level representation of a genre
Different genres within the same period	Rock compared with Pop within 1980–1984	Separation or overlap between genres during the same period
Different genres across different periods	Rock in 1980–1984 compared with Pop in 2000–2004	Not included in the main analysis because genre and temporal differences would be combined

Table 1: Scope of the optional track-level distribution-based comparisons.

The following Table 2 summarizes how each research question is connected to the corresponding methodology. All research questions are described in more detail afterwards.

2 Overview: Research Questions and Methodological Links

RQ	Question focus	Main methodology used to answer it
RQ1	Temporal genre evolution across modalities	Genre-level centroid drift measured using cosine and Euclidean distance; dataset-level temporal drift as a reference baseline; drift first averaged across temporal transitions within each genre and then compared across genres and modalities; coverage diagnostics; same-genre distributional drift across periods as an optional robustness extension
RQ2	Absolute and relative change in interpretable features	Raw feature trajectories for absolute change; within-period normalized trajectories for a genre’s position relative to the track-level feature distribution in the same period; raw dataset-level trajectories as reference; centroid-based genre-window summaries; selected Essentia and Lyrics TF-IDF features
RQ3	Temporal genre differentiation	Pairwise cosine similarity and Euclidean distance between genre centroids within the same period; global mean similarity and distance curves; coverage diagnostics; comparison of different genre distributions within the same period as an optional robustness extension
RQ4	Cross-modality similarity evolution	Comparison of centroid-based genre-pair similarity trajectories across modalities; correlations based on non-overlapping windows or methods accounting for temporal dependence; optional comparison of distribution-distance trajectories

Table 2: Connection between the research questions and the corresponding methodology. Dataset-level temporal drift provides the reference baseline, while centroid-based analyses constitute the main analysis. Distribution-based methods are optional extensions that examine whether centroid-based conclusions remain valid when within-genre variation is considered.

3 Research Questions

RQ1: Modality-Dependent Temporal Evolution

How much do music genres change over time across acoustic, visual, and lyrical feature modalities?

This question addresses whether temporal genre evolution is systematically stronger in some feature spaces than in others. The main analysis compares the distribution of genre-level average temporal drift across modalities using centroid-based summaries. This provides a transparent starting point for comparing temporal movement across MFCC, Essentia, MusicNN, VGG19, Word2Vec, and Lyrics TF-IDF.

Genre-level movement will be interpreted relative to dataset-level temporal drift. This helps distinguish genre-specific evolution from changes that affect the complete dataset or genre land-

scape. Cosine and Euclidean distance are used as complementary measures, as defined in Equations 9 and 10. Cosine distance captures changes in direction or feature profile, whereas Euclidean distance also captures the magnitude of displacement. The suitability of each distance measure will be checked for every feature representation.

The main cross-modality drift comparison will use standardized centroid representations. Within-period normalized centroids (Equation 8 in Section 4.7) describe changes in relative position, while raw or globally standardized representations (Section 4.4) are required to quantify general dataset drift across periods. Raw feature scales will not be compared directly across modalities.

Methodology link. RQ1 is answered by dataset-level baselines in Section 4.6, centroid-based temporal drift in Section 4.8, and average temporal drift comparison in Section 4.9. Distribution-based temporal drift in Section 4.13 is an optional robustness extension. The main outputs are dataset- and genre-level drift curves, genre-level drift distributions, modality-level summaries, and outlier genres.

Hypothetical intuitive result, not computed yet. A possible result could be that Lyrics TF-IDF shows greater average centroid drift than the acoustic modalities, suggesting stronger temporal movement in lyrical representations. If this movement is substantially larger than the general Lyrics TF-IDF dataset drift, this would provide evidence that the observed change is not explained solely by dataset-wide movement. This example is illustrative only and is not a computed result.

RQ2: Relative Genre Position and Interpretable Feature Change Within Time

How do selected interpretable acoustic and lyrical features change over time, both in their raw values and in a genre’s position relative to the track-level feature distribution in the same period?

This question focuses on interpretable feature modalities, especially Essentia and Lyrics TF-IDF. It combines two complementary perspectives on the same underlying features. First, it asks how a genre is positioned relative to the track-level feature distribution in the same temporal window. Second, it asks how this within-period relative position changes across temporal windows and how it relates to the corresponding raw feature trajectory.

For Essentia, this can be analyzed through interpretable audio descriptors such as loudness, energy, tempo, spectral features, or timbral features. For Lyrics TF-IDF, this can be analyzed through selected lyric terms or term groups. The goal is to identify whether a genre is above, below, or close to the period-level track average for a specific interpretable feature, and to track how this relative position develops over time alongside the genre’s raw, non-normalized feature trajectory and the corresponding dataset-level baseline.

The interpretable features will be selected before temporal hypothesis testing using pre-defined criteria: semantic interpretability, sufficient temporal coverage, low redundancy, and representation of distinct acoustic or lyrical dimensions. Any examples selected only after inspecting the results will be labelled explicitly as exploratory.

Methodology link. RQ2 is answered by within-period normalization in Section 4.5, dataset-level baselines in Section 4.6, centroid-based summary representation in Section 4.7, and raw/normalized

feature trajectories in Section 4.17. The centroid-based summary representation is used to aggregate track-level feature values at the genre-window level.

Hypothetical intuitive result, not computed yet. A possible result for Essentia could be that a genre becomes increasingly above the period-level track average in normalized loudness, while the raw dB baseline also increases across the dataset. This would allow the interpretation that the genre follows a general loudness trend but also changes relative to the contemporary track distribution. A possible result for Lyrics TF-IDF could be that a genre shows increasingly high normalized values for a selected lyric term relative to the period-level track average, while the raw TF-IDF trajectory helps clarify whether this reflects an absolute increase in term usage or mainly a relative shift. These examples are illustrative only and are not computed results.

RQ3: Temporal Genre Differentiation

Do music genres become more or less similar to one another over time across different feature modalities?

This question is addressed through pairwise genre similarity and distance within each modality and temporal window. The main analysis computes cosine similarity and Euclidean distance between genre centroids. This answers whether genres become more or less similar on average in terms of their average relative feature-space position and their total separation.

An optional distribution-based extension asks whether the same pattern is visible when genres are represented by their track-level distributions rather than single centroid vectors. This is relevant because centroids can move apart while distributions become wider and overlap more, or centroids can remain similar while internal distributions become more separated.

Since later periods may contain more genres and more genre pairs, similarity and separation trends must be interpreted together with coverage diagnostics.

Methodology link. RQ3 is answered by coverage diagnostics in Section 4.3 and pairwise centroid genre similarity and distance in Section 4.10. Distribution-level genre separation in Section 4.14 is an optional robustness extension. Global mean centroid similarity and distance curves will be interpreted together with coverage variables and results based on a stable set of genres.

Hypothetical intuitive result, not computed yet. A possible result could be that global mean centroid similarity decreases over time, suggesting that genre centroids become more separated. If distribution-level distances also increase, this would strengthen the interpretation that genres become more differentiated. If centroid similarity decreases but distributional overlap remains high, this would suggest that average genre positions diverge while the actual track-level distributions still overlap substantially. This example is illustrative only and is not a computed result.

RQ4: Cross-Modality Similarity Evolution

Do temporal changes in pairwise genre similarity follow similar patterns across feature modalities?

This question asks whether the similarity structure between genres changes in comparable ways across feature modalities. The main analysis does not compare feature values directly across modalities. Instead, it compares whether the similarity trajectory of the same genre pair evolves similarly across modalities.

This will primarily use centroid-based pairwise genre similarities because they are computationally efficient and comparable across all modalities. Correlations and significance tests will use non-overlapping windows or methods that explicitly account for temporal dependence. Spearman correlation will be used as the robust main correlation, with Pearson correlation as an optional complementary measure. If suitable distribution-level distances are obtained, the same idea may be applied to genre-pair distance trajectories as an optional extension.

The primary comparisons will be within related modality families, such as MFCC vs. Essentia for audio-based features or Word2Vec vs. Lyrics TF-IDF for lyric-based features. Cross-family comparisons, such as audio vs. lyrics, will be treated as exploratory and interpreted as temporal co-evolution of genre relationships rather than direct feature similarity.

Methodology link. RQ4 is answered by cross-modality similarity trajectories in Section 4.11. Optional robustness checks use distribution-based distance trajectories from Section 4.15.

Hypothetical intuitive result, not computed yet. A possible result could be that MFCC and Essentia show high positive correlations in pairwise centroid-similarity trajectories, indicating similar acoustic genre-relationship dynamics. If distribution-distance trajectories show a similar result, the finding would be more robust. Audio vs. lyrics may show weaker correlations, suggesting that acoustic and lyrical genre relationships can evolve independently. This example is illustrative only and is not a computed result.

4 Methodology

4.1 Dataset and Feature Modalities

The analysis is based on the Music4ALL dataset. Tracks are represented using six feature modalities:

Modality	Feature type	Role in analysis
MFCC	continuous acoustic descriptors	acoustic/timbral baseline
Essentia	interpretable audio descriptors	acoustic interpretation
MusicNN	learned audio representation	learned audio embedding
VGG19	learned visual representation	visual/cover-art representation
Word2Vec	lyric semantic embedding	semantic lyric representation
Lyrics TF-IDF	sparse lyric text vector	interpretable lyric-term representation

Table 3: Feature modalities used in the adapted thesis pipeline.

Each modality is analyzed separately because the modalities have different dimensionalities, value distributions, and semantic interpretations.

4.2 Temporal Windowing

The adapted analysis will use sliding five-year windows with a one-year hop size.

1955–1959, 1956–1960, 1957–1961, ...

The default setup is:

- window length: 5 years,
- hop size / step size: 1 year,
- minimum tracks per genre-window for centroid analysis: 20,
- higher thresholds for optional distribution-based modeling: at least 30,
- minimum valid windows per genre: 10,

Sliding windows provide a smoother temporal view than fixed non-overlapping five-year periods. Because adjacent windows share four of five years, neighboring estimates are not independent. Sliding windows with a one-year hop will therefore be used for descriptive trajectories. Correlations and inferential statistical tests will primarily use non-overlapping five-year windows with a five-year hop. More advanced alternatives that explicitly account for autocorrelation, such as block bootstrap procedures or time-series models, may be considered where necessary.

4.3 Coverage Diagnostics

Before interpreting drift, similarity, or distributional change, the thesis will inspect the available data coverage in each temporal window.

For every modality and window, the following values will be computed:

- number of tracks,
- number of genres,
- number of valid genres,
- distribution of the number of genres per five-year interval,
- number of valid genre pairs,
- number of tracks per valid genre,
- proportion of tracks contributed by each genre,
- proportion of missing features,
- number of genre-windows eligible for centroid analysis,
- number of genre-windows eligible for optional empirical track-level analysis,
- number of genre-windows eligible for optional model-based distribution analysis,
- number of valid distribution-level genre-pair comparisons.

Coverage diagnostics are needed because temporal patterns may be influenced by changing genre availability.

4.4 Data Representations Used in the Analysis

The pipeline will save and use five related representations:

1. **Raw/original track-level feature values:** values in the original feature scale, such as loudness in dB or tempo.
2. **Within-period normalized track-level values:** z-standardized values computed separately within each modality and temporal window.
3. **Globally standardized track-level values:** z-standardized values based on the complete temporal analysis sample of a modality, preserving dataset-level temporal movement on a common scale.
4. **Centroid summaries:** one average vector per genre, modality, and temporal window.
5. **Distributional genre-window representations:** optional empirical or model-based representations of the actual track-level datapoints inside a genre-window.

For globally standardized values, one mean and standard deviation are calculated across the complete temporal analysis sample of a modality and then used for every temporal window:

$$x_{i,j}^{(m,\text{global})} = \frac{x_{i,j}^{(m)} - \mu_j^{(m,\text{global})}}{\sigma_j^{(m,\text{global})}}. \quad (1)$$

Because the same reference is used for every period, general temporal movement remains visible.

The role of these representations is different:

Representation	Used for	Interpretation
Raw/original track-level values	Interpretable feature trajectories, captions, selected genre examples, dataset-level baselines	Actual feature values in the original scale, where interpretable
Within-period normalized track-level values	Relative genre position, standardized feature comparison, and optional track-level analysis	Relative position compared with the period-level track distribution
Globally standardized track-level values	Dataset-level temporal drift and complementary genre-level drift	Temporal movement on a fixed modality-specific scale
Centroid summaries	Main drift analysis, pairwise genre similarity and distance, selected genre trajectories	Average genre position in a modality and window
Distributional representations	Optional empirical or model-based comparisons	Track-level genre-window structure including mean, spread, covariance, overlap, and possible subclusters

Table 4: Role of raw values, normalized values, centroids, and distributional representations in the adapted analysis.

4.5 Normalization Strategy

The main normalization strategy will be **within-period normalization**. This means that z-standardization is performed separately for each modality and temporal window, but not separately for each genre.

For track i , feature dimension j , modality m , and temporal window t , the normalized value is:

$$z_{i,j}^{(m,t)} = \frac{x_{i,j}^{(m)} - \mu_j^{(m,t)}}{\sigma_j^{(m,t)}} \quad (2)$$

where $\mu_j^{(m,t)}$ and $\sigma_j^{(m,t)}$ are computed across all valid tracks within modality m and temporal window t . The same normalization parameters are therefore applied to tracks from all genres in that window. After normalization, tracks are grouped by genre and averaged to obtain genre-level centroids. Because all genres use the same reference, their centroids remain comparable within the same temporal window.

This means that normalized values answer:

How is a track or genre positioned relative to the track-level feature distribution in the same temporal window?

They do not directly answer:

How did the raw feature value change across the full historical dataset?

Within-period normalization does not preserve general dataset movement because the mean of all normalized tracks in every window is zero by construction. Raw or globally standardized values are therefore required for dataset-level drift and absolute temporal change. Raw feature trajectories will be shown alongside within-period normalized trajectories for selected interpretable features in RQ2.

The primary reference is the track-weighted period distribution. As a sensitivity analysis, a genre-balanced reference will first average each feature within every valid genre and then give all valid genres equal weight. This tests whether conclusions are dominated by genres with larger track counts.

4.6 Dataset-Level Baselines

Dataset-level baselines describe how the complete dataset changes over time and provide a reference for interpreting genre-level movement. Let $x_i^{(m,r)}$ denote the feature vector of track i in modality m , using representation r , where r refers to raw or globally standardized values.

For modality m , representation r , and temporal window t , the track-weighted dataset centroid is:

$$c_t^{(m,r,\text{dataset})} = \frac{1}{N_t^{(m)}} \sum_{i \in T_t^{(m)}} x_i^{(m,r)}, \quad (3)$$

where $T_t^{(m)}$ is the set of all valid tracks and $N_t^{(m)}$ is their number. This centroid represents the average track in the temporal window. Because genres with more tracks contribute more strongly, a genre-balanced centroid will be used as a sensitivity analysis:

$$c_t^{(m,r,\text{genres})} = \frac{1}{|\mathcal{G}_t^{(m)}|} \sum_{g \in \mathcal{G}_t^{(m)}} c_{g,t}^{(m,r)}, \quad (4)$$

where $\mathcal{G}_t^{(m)}$ is the set of valid genres. The track-weighted centroid represents the average track, while the genre-balanced centroid represents the average genre.

For an interpretable raw feature j , the dataset-level raw baseline is:

$$B_{j,t}^{(m,\text{raw})} = \frac{1}{N_t^{(m)}} \sum_{i \in T_t^{(m)}} x_{i,j}^{(m)}. \quad (5)$$

This baseline answers how the average raw feature value of all tracks changes over time. For selected interpretable features in RQ2, the genre’s raw trajectory, the raw dataset-level trajectory, and the genre’s within-period normalized trajectory will be shown together.

For two temporal windows t_a and t_b , dataset-level temporal drift is:

$$D_{t_a,t_b}^{(m,r,\text{dataset})} = d\left(c_{t_a}^{(m,r,\text{dataset})}, c_{t_b}^{(m,r,\text{dataset})}\right), \quad (6)$$

where d is cosine or Euclidean distance. Genre-level and dataset-level drift will be calculated using the same representation and distance measure and interpreted together rather than directly subtracted. Within-period normalized values will not be used for track-weighted dataset drift because their dataset centroid is approximately zero in every window by construction.

Dataset- and genre-level drift will first be calculated separately within each modality. Raw

feature values will not be compared directly across modalities because scales and dimensionalities differ. Cross-modality analyses will instead compare drift patterns and summary measures derived from appropriate standardized representations.

4.7 Centroid-Based Summary Representation

For each genre g , modality m , and temporal window t , the centroid is computed by averaging the feature vectors of all tracks belonging to that genre in that window. Let $T_{g,t}^{(m)}$ denote these tracks and $N_{g,t}^{(m)}$ their number.

For raw features:

$$c_{g,t}^{(m,\text{raw})} = \frac{1}{N_{g,t}^{(m)}} \sum_{i \in T_{g,t}^{(m)}} x_i^{(m)} \quad (7)$$

For within-period normalized features:

$$c_{g,t}^{(m,\text{wpz})} = \frac{1}{N_{g,t}^{(m)}} \sum_{i \in T_{g,t}^{(m)}} z_i^{(m,t)} \quad (8)$$

The normalization parameters used to obtain $z_i^{(m,t)}$ are calculated across all valid tracks in the modality and temporal window, not separately for each genre. Therefore, the centroid across all normalized tracks is approximately zero, but individual genre centroids are not forced to zero.

The raw centroid represents the average feature vector of a genre-window in the original feature space. The within-period normalized centroid represents the average position of the genre relative to the period-level track distribution.

Centroids are used as the **main summary representation**. They are not treated as the only possible representation of a genre. Their role is to provide a clear first approximation of temporal movement and genre similarity. Optional empirical spread and distribution-based methods can later test whether this summary is sufficient.

4.8 Centroid-Based Temporal Drift

Centroid-based temporal drift measures how much the centroid of the same genre changes between two temporal windows. Dataset-level drift from Section 4.6 is used as a reference baseline to assess whether genre-level movement follows a general dataset-wide trend.

For genre g , modality m , and two temporal windows t_a and t_b , cosine drift is:

$$D_{g,t_a,t_b}^{(m,\text{cos})} = 1 - \frac{c_{g,t_a}^{(m)} \cdot c_{g,t_b}^{(m)}}{\|c_{g,t_a}^{(m)}\| \|c_{g,t_b}^{(m)}\|} \quad (9)$$

and Euclidean drift is:

$$D_{g,t_a,t_b}^{(m,\text{L2})} = \left\| c_{g,t_a}^{(m)} - c_{g,t_b}^{(m)} \right\|_2. \quad (10)$$

Cosine distance captures changes in direction or feature profile, whereas Euclidean distance also captures the magnitude of displacement. The suitability of each measure will be checked separately for every modality, considering whether the representation was designed or optimized

for cosine, Euclidean, or another distance. Raw Euclidean distances will not be compared directly across modalities because scales and dimensionalities differ.

The interpretation depends on the representation. Raw or globally standardized centroids describe movement on a fixed temporal scale. Within-period normalized centroids describe changes in a genre’s position relative to the track distribution in the same period. Genre-level drift based on raw or globally standardized representations will be interpreted together with dataset-level drift calculated using the same representation and distance measure. The values will be compared as contextual measures rather than directly subtracted.

Possible extensions include long-range drift variants:

- neighboring-window drift,
- multi-lag drift, for example 5-year, 10-year, and 20-year gaps,
- baseline-to-window drift, for example distance from 1970 to later windows,
- all-pairs same-genre cross-period drift.

Local movement and net displacement will not be treated as equivalent. Directionality is defined as the direct distance between the first and last centroid divided by the sum of consecutive movements:

$$\text{Directionality}_g^{(m)} = \frac{d(c_{g,t_{\text{first}}}^{(m)}, c_{g,t_{\text{last}}}^{(m)})}{\sum_t d(c_{g,t}^{(m)}, c_{g,t+1}^{(m)})}. \quad (11)$$

This separation prevents local statistical fluctuations from being interpreted as net historical change.

4.9 Average Temporal Drift Comparison

After computing genre-level centroid drift, the analysis will aggregate drift values to compare average temporal movement across modalities.

For each genre g , modality m , and distance measure d , the mean neighboring-window centroid drift is:

$$\bar{D}_g^{(m,d)} = \frac{1}{|\mathcal{E}_g^{(m)}|} \sum_{(t_a, t_b) \in \mathcal{E}_g^{(m)}} D_{g,t_a,t_b}^{(m,d)}, \quad (12)$$

where $\mathcal{E}_g^{(m)}$ is the set of valid neighboring-window comparisons for genre g . Drift is averaged over time within each genre first. Modality summaries are then computed from the distribution of these genre-level means so that genres with more valid windows do not automatically receive greater weight.

Because distances are non-negative, random variation in estimated centroids can produce positive drift even if the underlying representation is stable. Observed drift will therefore be interpreted together with bootstrap confidence intervals or a resampling-based null reference. Important comparisons will also be repeated using non-overlapping windows. Mean neighboring-window drift measures average local movement and will not be interpreted as net historical change; direct displacement and directionality will be reported separately.

The main outputs are distribution of genre-level mean centroid drift values by modality, modality-level mean and median drift values, outlier genres, temporal curves of average centroid drift per modality, and statistical tests for modality differences where appropriate.

4.10 Pairwise Genre Similarity

Pairwise genre similarity and distance are computed within the same modality and the same temporal window.

For genres g and h , modality m , and window t :

$$S_{g,h,t}^{(m,\text{centroid})} = \frac{c_{g,t}^{(m)} \cdot c_{h,t}^{(m)}}{\|c_{g,t}^{(m)}\| \|c_{h,t}^{(m)}\|} \quad (13)$$

The corresponding Euclidean distance is:

$$\Delta_{g,h,t}^{(m,\text{L2})} = \left\| c_{g,t}^{(m)} - c_{h,t}^{(m)} \right\|_2. \quad (14)$$

This means that genre similarity asks:

How similar are two genres within the same temporal window and modality according to their average relative positions?

For each modality and temporal window, the global mean genre similarity is computed by averaging over all valid genre pairs:

$$\bar{S}_t^{(m,\text{centroid})} = \frac{1}{|\mathcal{P}_t^{(m)}|} \sum_{(g,h) \in \mathcal{P}_t^{(m)}} S_{g,h,t}^{(m,\text{centroid})} \quad (15)$$

where $\mathcal{P}_t^{(m)}$ is the set of all valid genre pairs in modality m and temporal window t .

Because similarity and distance depend on dimensionality and feature distribution, observed values will be compared with a null reference created by permuting genre labels among tracks within the same modality and temporal window and recomputing the centroids and pairwise statistics. This preserves dimensionality, feature distribution, genre sizes, and coverage while removing meaningful genre structure.

A decreasing cosine similarity or increasing Euclidean distance suggests greater genre separation only if the pattern also differs from the null reference. Results will additionally be interpreted together with coverage diagnostics, a stable-genre-set sensitivity analysis, and optional distribution-level separation.

4.11 Cross-Modality Correlation of Similarity Trajectories

To inspect whether modalities evolve similarly or differently, the thesis will compute correlations between pairwise genre-similarity trajectories across modalities.

For a genre pair (g, h) , modality m , and temporal window t , the pairwise centroid similarity is $S_{g,h,t}^{(m,\text{centroid})}$. For two modalities m_1 and m_2 , the similarity trajectories for the same genre pair can be compared across temporal windows:

$$\rho_{g,h}^{(m_1,m_2)} = \text{corr} \left(\{S_{g,h,t}^{(m_1,\text{centroid})}\}_t, \{S_{g,h,t}^{(m_2,\text{centroid})}\}_t \right) \quad (16)$$

The main analysis is computed for pairs of genre centroids rather than pairs of individual tracks because RQ4 concerns genre-level relationships. It does not compare feature values directly across modalities; it tests whether the relationship between the same two genres follows a similar temporal pattern in both modalities. Optional track-level comparisons may later test whether centroid-based conclusions remain valid when variation between individual tracks is considered.

4.12 Optional Track-Level Distribution Analysis

If the centroid-based results are relevant and stable, optional track-level analyses will examine whether they remain valid when variation between individual tracks is considered.

For a genre g , modality m , and temporal window t , let

$$X_{g,t}^{(m)} = \{z_i^{(m,t)} : i \in T_{g,t}^{(m)}\} \quad (17)$$

denote the set of normalized track-level feature vectors in the genre-window. The analysis will begin with empirical measures such as average distance to the centroid, pairwise track similarity, feature variance, and covariance trace. These measures do not require a probability model.

If comparison of complete genre-window distributions provides useful additional information, empirical or model-based distribution distances may be considered. The method will be selected based on the observed distribution structure, feature dimensionality, computational feasibility, and available track coverage.

4.13 Optional Distribution-Based Temporal Drift

Distribution-based temporal drift compares the track-level representation of the same genre across periods. Its purpose is to examine whether centroid-based drift remains valid when changes in within-genre variation are considered.

Empirical measures such as variance, covariance trace, pairwise track distance, and average distance to the centroid will be examined first. If a comparison of complete distributions is useful, empirical Wasserstein or Earth Mover’s Distance may be considered. Because these methods can be expensive and unstable in high dimensions with limited coverage, a Gaussian approximation may be used when appropriate:

$$X_{g,t}^{(m)} \sim \mathcal{N}(\mu_{g,t}^{(m)}, \Sigma_{g,t}^{(m)}). \quad (18)$$

The Gaussian approximation is not required for empirical descriptors. It is only a possible compact representation that allows Wasserstein-2 distance to compare changes in both mean and covariance. Distribution-based results will be compared with centroid-based results to identify cases in which centroid movement and within-genre variation lead to different conclusions.

4.14 Optional Distribution-Level Genre Separation

If distribution-level genre separation provides useful additional information, the track-level distributions of two different genres will be compared within the same modality and temporal window. Empirical distance measures will be considered first. Gaussian Wasserstein distance may be used only when a Gaussian approximation is appropriate and sufficiently stable.

This answers:

How far apart are two genre distributions within the same modality and temporal window?

The resulting distances provide an optional distribution-level analogue to the centroid-similarity analysis. Agreement would support the centroid-based conclusion, while disagreement could indicate that changes in internal spread or overlap affect the interpretation.

4.15 Distribution-Based Distance Trajectories Across Modalities

If suitable distribution-level genre-pair distances are obtained, their temporal trajectories may be compared across modalities as an optional robustness analysis for RQ4. The same coverage, temporal-dependence, and multiple-comparison requirements used in the centroid-based analysis will apply. No specific distribution model is selected in advance.

4.16 Feature Stability and Spread Measures

In addition to drift and similarity, the thesis will compute clearly separated measures of internal variation for genre-window representations.

The purpose of these measures is to quantify how compact or internally variable a genre is within a temporal window. This is important because two genre-windows can have similar centroids but very different internal spread. The measures will not be merged into a single ambiguous stability score.

Depending on the feature representation, stability can be measured in different ways:

- compactness: average pairwise song similarity within the same genre-window,
- average feature variance across dimensions,
- covariance spread: trace of the empirical or model-based covariance matrix,
- internal spread: average distance to the genre centroid,
- optional distributional overlap estimates between selected genre pairs.

For track-level representations, a compactness measure is the average pairwise similarity among tracks within the same genre-window:

$$\text{Compactness}_{g,t}^{(m)} = \frac{1}{|\mathcal{Q}_{g,t}^{(m)}|} \sum_{(i,k) \in \mathcal{Q}_{g,t}^{(m)}} \text{sim}(z_i^{(m,t)}, z_k^{(m,t)})$$

where $\mathcal{Q}_{g,t}^{(m)}$ is the set of all valid track pairs within genre g , modality m , and window t . If the number of pairs is computationally prohibitive, a fixed-size random sample of track pairs will be used.

An empirical or Gaussian-model-based covariance spread measure is:

$$\text{Spread}_{g,t}^{(m)} = \text{Tr} \left(\Sigma_{g,t}^{(m)} \right)$$

Lower spread indicates a more compact genre representation. Higher spread indicates a more internally diverse genre representation.

These internal-variation measures will be used to interpret centroid-based and optional distribution-based results.

4.17 Raw and Normalized Feature Trajectories for Selected Genres

For selected interpretable examples, especially in Essentia and Lyrics TF-IDF, the analysis will show both within-period normalized values and raw or original feature values.

Features will be selected before temporal hypothesis testing using four criteria: semantic interpretability, sufficient temporal coverage, low redundancy with already selected features, and representation of distinct acoustic or lyrical dimensions. For Essentia, the selection will cover categories such as dynamics or loudness, rhythm or tempo, spectral characteristics, timbre, and tonal or harmonic characteristics. For Lyrics TF-IDF, sufficiently frequent individual terms, pre-defined term groups, or transparently derived term clusters may be used. Any examples selected after inspecting temporal results will be labelled as exploratory rather than confirmatory.

The normalized trajectory is used to answer how a selected genre is positioned relative to the period-level track distribution. For example, a positive normalized loudness value means that the genre is, on average, louder than the average track in the same window. A negative value means that it is below the period-level track average.

However, normalized values are not always directly interpretable in the original unit of the feature. Therefore, whenever a normalized trajectory is shown for an interpretable feature, the thesis will also provide the corresponding raw-value interpretation where possible.

For genre-specific visualizations, the thesis will include the track-weighted raw dataset baseline. A genre-balanced baseline may additionally be shown as a sensitivity analysis.

The visualization will compare:

- the selected genre’s raw feature trajectory in the original feature scale,
- the window-level raw baseline,
- the selected genre’s within-period normalized trajectory,
- notes or captions translating normalized values back to original units where the feature is directly interpretable,
- optionally the genre’s rank or percentile among all valid genres in the same window.